NNN-O-001191 (DSA-DM)
4 August 1966
Interim Revision of
Fed. Spec. W-O-830b (in Part)
(See 6.4)

# INTERIM FEDERAL SPECIFICATION

## OVEN, LABORATORY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification W-O-830b.

### 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric gravity convection-type laboratory ovens suitable for operation on 220 volts, 60 cycles, alternating current, in the temperature range from 35° Centigrade to 260° Centigrade.

1.2 Classification.

1.2.1 Styles and sizes. The ovens shall be of the following styles and sizes as specified (see 6.2).

1.2.1.1 Styles. Style 1. Single door.
Style 2. Double door.

FSC 6640

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1.2.1.2 Sizes. The ovens covered by this specification shall be of the sizes specified in table I. The dimensions shown in table I shall be considered as the clear working space within the oven.

Table I - Inside dimensions

| Size | Width Inches | Depth Inches | Height Inches |
|---|---|---|---|
| 1 | 13 | 14 | 13 |
| 2 | 19 | 14 | 19 |
| 3 | 37 | 19 | 25 |

2. APPLICABLE SPECIFICATIONS AND STANDARDS AND OTHER PUBLICATIONS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| GG-T-336 | Thermometers, Self-Indicating, Liquid-In Glass, Laboratory. |
| UU-P-271 | Paper, Wrapping, Waterproofed Kraft. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

STANDARDS

Federal

| | |
|---|---|
| Fed Std. No. 66 | Steel: Chemical Composition and Hardenability. |
| Fed Std. No. 102 | Preservation, Packaging, and Packing Levels. |
| Fed Std. No. 123 | Marking for Domestic Shipment (Civilian Agencies) |



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Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-130 | Identification Marking of U.S. Military Property. |

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., Standards:

| | |
|---|---|
| UL 50 | Standard for cabinets and boxes |
| UL 67 | Standard for Panel boards. |
| UL 499 | Standard for Heating Appliances, Electric |

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y., 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California, 95050).



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"National Fire Protection Association:

NFPA No. 70 National Electrical Code.

"(Application for copies should be addressed to the National Fire protection Association, 85 John Street, New York 38, N.Y., 222 West Adams Street, Chicago 6, Ill., and 465 California Street, San Francisco 4, Calif.)"

"American Standards Association, Inc.:

ASA C73.20 - 1963 American Standard for Outlet Receptacles, Attachment Plug Caps and Appliance Plugs.

"(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N.Y. 10016.)"

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and as specified herein.

3.1.1 Corrosion-resisting steel. Corrosion-resisting steel shall conform to Federal Standard No. 66, AISI 304 for plates, sheets, strips, and structural shapes, and AISI 303 for bars and forgings. Where welding is limited to spot welding, plates, strips, sheets, and structural shapes may conform to AISI 302. The chemical composition shall conform to the requirements of Federal Standard No. 66.

3.1.2 Insulation. Insulation material shall be noncombustible, nonhygroscopic, and nonpowdering, made from asbestos air cell or glasswool having a density of not less than 2.0 pounds per cubic foot. As installed, the insulation shall be precompressed or bonded together with a high-heat-resistant material, to form a blanket not less than 3 inches thick.

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit to the contracting agency, prior to delivery , proof that the ovens he will supply under this specification conform to the requirements of the Underwriters' Laboratories, Inc. Standards Nos. UL 50, UL-67 and UL 499. Such proof shall consist of a laboratory report or listing by the Underwriters' Laboratories, Inc. The label or listing of the Underwriters Laboratories shall be acceptable as evidence that equipment conforms to the above. A copy of each laboratory report shall be furnished to the contracting officer.



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3.2.2 In lieu of the label or listing by the Underwriters' Laboratories, Inc., the contractor may submit a laboratory report (see 6.5) by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit a laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

3.2.3 Compliance with Underwriters' Laboratories, Inc., requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.3 Design. The oven shall be a rectangular cabinet of double-wall construction, with thermal-insulating material between the walls to conserve heat. Each oven cabinet shall be designed with shelf supports in the working chamber to hold shelves at various heights. In addition to the repair parts (see 3.8), each oven shall be equipped with two shelves, a thermometer, indicator light, variable thermostat control, switch or switches under thermostatic control for controlling power input to the heaters, and suitable high-resistance alloy heating elements. The ovens shall be supported on legs to permit installation on laboratory benches or have modified legs for floor mounting. All electrical wiring and control circuits shall be enclosed in a wiring compartment thermally insulated from the working chamber. The ovens shall be designed to operate on 220 volts, 60 cycles, single phase, alternating current. The operating temperature range of the ovens shall be from 35 degrees Centigrade to 260 degrees Centigrade. Each oven shall have two air vents, one fresh air intake vent and one exhaust vent. The vents shall be designed so that heat transfer by convection shall set up air flow patterns for heat distribution within the working chamber conducive to accurate temperature control.

3.4 Details of components.

3.4.1 Cabinet. The cabinet shall be of double-walled metal construction, except walls of control compartments not adjoining the inner wall defining the oven working space, mounted on a steel framework to prevent distortion. The assembly of the cabinet shall be either all welded or fabricated with sheet-metal screws, bolts, and locknuts or a combination of both. The cabinet shall provide space for all wiring, heaters, and controls. Exposure of insulation and open seams in the interior or exterior of the cabinet shall not be permitted.



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3.4.1.1 Framework. The framework shall be of steel and of all-welded construction.

3.4.1.2 Inner walls. The inner walls, including the top and bottom, defining the oven working space shall be made of corrosion-resisting steel not less than 20 gauge (0.0375 inch, nominal). There shall also be an interior barrier wall made of carbon steel not less than 20 gage (0.0359 inch, nominal) to support the insulation separating the control and wiring compartment from the oven working space.

3.4.1.3 Exterior walls. The exterior walls of the oven shall be corrosion-resisting steel not less than 18 gage (0.050). Construction of any seams shall cause no buckling or bulging after fabrication.

3.4.1.4 Structural bracing. The inner and outer walls shall be braced on the unexposed face, as required by formed corrosion-resisting sheet-steel struts to prevent buckling and bulging. The struts shall be noncontinuous from inner to outer walls. Reinforcing shall be provided at points of attachment of hinges and other load supporting fittings.

3.4.1.5 Air vents. Each oven shall contain an air intake vent located on the bottom of the cabinet and an exhaust vent located on the top of the cabinet. The air intake vent shall be a perforated diffusion panel for the passage of air into the bottom of the cabinet. The location, size and design of the exhaust vent shall be consistent with good practice in the industry and the performance requirements of this specification. Both vents shall be made of corrosion-resisting steel or brass and may be cast or fabricated from sheet stock. Brass, when used, shall be chromium-plated.

3.4.1.6 Thermometer opening. A suitable opening in the top of the oven shall be provided which will hold the thermometer snugly and not permit the entrance of air into the oven. The opening may be integral with the ventilator assembly.

3.4.1.7 Shelf supports. Shelf supports of corrosion-resisting steel shall be attached to the right and left walls of the interior of the oven. The supports shall permit the shelves to be placed at intervals of not over 1-1/2 inches for all size ovens from top to bottom of the working chamber, the top most position to be 3-1/2 inches from the top. Shelf supports shall be designed so as to allow unloaded shelves to be extracted 1/2 the depth of the shelf and still remain intact.



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3.4.1.8 Legs. The cabinet shall be supported by a minimum of 4 legs of corrosion-resisting steel, integral with, or either welded or bolted, with 4 bolts for each leg, to the frame. The legs may be covered by a corrosion-resisting steel skirt provided there is sufficient open area for the circulation of air under the cabinet to prevent excessive temperature on the supporting surface.

3.4.1.9 Thermal insulation installation. Thermal insulation as described in 3.1.2 shall be placed between the inner and outer walls of the cabinet except where wiring or control compartments adjoin the inner working chamber of the oven. The insulation shall be placed between the inner wall of the oven and the interior barrier wall of the compartment provided for in 3.4.1.2. The insulation shall be supported in such a manner that the minimum amount of continuous metal consistent with strength and rigidity shall pass through the double walls of the cabinet. The insulation shall not support any load in excess of its own weight and shall not come in contact with uninsulated electrical conductors. Insulation shall be walled off by corrosion-resisting steel or other suitable corrosion-resisting materials where openings from the inner to outer walls are necessary.

3.4.2 Doors. Oven doors shall be of double-walled corrosion-resisting steel construction the same as the cabinet and shall be close fitting within the door opening to prevent heat loss. The door shall have not less than a 1-inch overlap around the door opening and shall seat tightly against a suitable resilient heat resistant seal around the edge to prevent air leaks. The space between inner and outer faces of the door shall be thermally-insulated in accordance with 3.1.2 and 3.4.1.9. Doors shall be reinforced at attachment points of hinges, handles, and latches. The doors shall be mounted on hinges so as to provide an unobstructed entrance to the oven for the full width of the inner wall, and designed so as not to strike the controls when opened.

3.4.2.1 Style 1. Style 1 ovens shall have a single door extending the full width and height of the working chamber. The door shall be mounted on at least two hinges and the hinges shall be on the left side of the oven as viewed from the front.

3.4.2.2 Style 2. Style 2 ovens shall have two doors hung on at least two hinges each. The doors shall be hung one on each side of the cabinet door opening and in the closed position the two doors together shall extend over the full width and height of the working chamber. There shall be no center post in the door opening.



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3.4.3 Shelves. Each oven shall be equipped with two removable expanded mesh-type corrosion-resisting steel shelves. The shelves shall be approximately the same size as the width and depth of the oven working chamber.

3.4.4 Thermometer. Thermometers furnished with each oven shall be type II, conforming to GG-T-336 with the following exceptions: The thermometer shall be graduated from -10 to 360° C in subdivisions of 1° C. The length shall be 400 ≠ 6 mm with an accuracy to within plus or minus 2.5° C up to 300° C and plus or minus 5.0° C from 300° C to 360° C. The gas used in the thermometer shall not contain any oxygen. The diameter at bore shall be not less than 0.10 mm and shall be approximately uniform throughout the length of the calibration. An expansion chamber shall be provided at the upper end of the bore of a capacity at least double that of the capillary tube. Width of the graduation lines shall not exceed 0.2 mm. Coloring material in graduations, numerals and letters shall be durable.

3.4.5 Card holder. A card holder shall be located and attached on the face of the cabinet so that there is no interference with the opening and closing of the doors. The card holders shall accommodate a 3 by 5 inch card with the long axis horizontal. The holder shall permit the card to be easily inserted and removed and shall have an exposed card area of not less than 2-3/4 by 4-3/4 inches.

3.4.6 Hardware. Door hinges, latches, and all other oven fittings, except door handles, shall be brass or high-pressure cast zinc alloy chromium over nickel plated, and highly polished. Door handles shall be thermosetting plastic or other suitable heat-resistant material. Handles shall not become overheated or be injurious to personnel when the oven is operating at maximum temperature.

3.4.7 Finish. The interior and exterior surfaces of the oven shall be made of corrosion-resisting steel. Exposed surfaces shall be finished in No. 4 commercial grind polish.

3.4.8 Electrical components. All electrical components of the oven, including electrical insulation, shall meet the requirements of the National Fire Protection Association, NFPA No. 70 and shall be suitable for the intended use, including adequate resistance to heat and moisture absorption.



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3.4.8.1 Heating units. Oven-heating units shall be designed to operate on 220 volts, 60 cycles, single phase alternating current and shall be capable of producing and maintaining oven temperatures from 35 degrees Centigrade to 260 degrees Centigrade. The heating unit elements shall be either exposed ribbon or round resistance wire, wound on a suitable grid structure constructed so that no unreasonable sagging or permanent distortion of the wire or structure shall result from normal oven operation; or the resistance wire shall be embedded in a suitable refractory or insulating material held in place by a corrosion-resisting metal sheath. The assembly shall not become grounded to the sheath when subjected to the shocks and vibrations of normal oven use.

3.4.8.1.1 Power density. The heating unit elements shall have a power density of not more than 8 watts per square inch of effective heating surface.

3.4.8.1.2 Mounting. Heating units shall be provided with means to prevent injury to personnel handling material in the oven during operation. The heating elements shall be protected against excessive dripping of liquids on either the exposed element wire or element sheath.

3.4.8.1.3 Terminals. The terminals of the heating unit shall be conveniently accessible for wiring test, and replacement purposes.

3.4.8.2 Heating Unit switch. Power input to the heating unit shall be controlled by a multiposition switch or other input controller, which shall be separate from the thermostat control to permit operation of the oven without an overshoot of the temperature value for which the thermostat is set, within the range given in 3.2. Switch positions shall be permanently and plainly marked.

3.4.8.3 Thermostat. The thermostat shall be in series with the switch and the heating element and shall be fully enclosed to protect the instrument against dust, dirt, moisture, or corrosive media. No transformer or auxiliary current supply shall be required and the thermostat shall be suitable for operation with the power supply specified in 3.4.8.1. Adjustment of the thermostat shall be by means of a knob and graduated reference dial allowing for a variation in temperature within the oven from 35 degrees C to 260 degrees C. The thermostat shall be capable of automatically maintaining the oven temperature as set to within the tolerance specified in 3.6.1 . The thermostat sensing element shall be suitably located inside the working chamber of the oven to insure sensitive response to temperature fluctuations.



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3.4.8.4 Indicator light. The oven shall be equipped with an automatic indicator light, showing red when energized, connected between the heating elements and the switch. The light shall signal automatic temperature control by the thermostat by indicating the make and break of the controlled heater circuit. The light shall be "ON" when the heater elements are energized. The indicator lamp shall be installed so as to permit replacement from the exterior of the oven. The indicator light shall be installed on the front surface of the cabinet. The lamp shall be a standard, incandescent, 115 volt, 6 watt bulb, shall have a candelabra-screw base and be a class B lamp, with a C-7A filament with an average laboratory life of 1500 hours.

3.4.8.5 Circuit wiring. The circuit wiring of the oven shall be completely enclosed in a control and wiring compartment integral with the cabinet. All exposed wiring shall be covered with suitably impregnated asbestos, fibrous glass sleeving or other suitable heat and moisture resisting insulation. Electrical connections shall not be adversely affected by the heat produced as a result of normal use of the oven. Circuit wiring shall terminate at a suitable terminal block for connection to the power supply.

3.4.8.6 Supply connection.

3.4.8.6.1 Armored cable and conduit connection. Provisions shall be made for the connection of armored cable or conduit for size 3 ovens.

3.4.8.6.2 Extention cords. Sizes 1 and 2 ovens shall be provided with an attached 6-foot flexible cord and attachment plug cap. Cord shall be three-wire, and plug shall be grounded type suitable for use with receptacles specified in ASA C73.20-1963 for caps and receptacles, 250 volts, 15 amperes, 2 poles, 3 wire, grounding type, general purpose.

3.5 Dimensions.

3.5.1 Oven interior. The minimum inside dimensions for each size oven shall be as specified in 1.2.1.2 and shall be determined from the clear working space inside the oven with the door or doors closed.

3.5.2 Oven exterior. The overall dimensions of the oven shall be as small as possible consistent with the specified interior dimensions and with the other requirements of this specification.

3.6 Performance.

3.6.1 Shelf deflection. Shelves shall not deflect more than 1/4 inch or receive any permanent deformation when tested as specified in 4.4.1.



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3.6.2 Shelf support rigidity. Shelf supports shall not buckle or bend on the surfaces to which they are attached when tested as specified in 4.4.2.

3.6.3 Thermometer. The thermometer shall be tested for accuracy and durability of coloring material as specified in 4.4.3 for conformance to 3.4.4.

3.6.4 Operating requirements. The oven shall operate satisfactorily on every respect for a period of 24 hours, when tested as specified in 4.4.4.

3.6.5 Heating time. The maximum operating temperature of 260 degrees C. shall be attained within 50 minutes from the time the power is applied to the heating element, when tested as specified in 4.4.5.

3.6.6 Heat loss. The thermal insulation shall be sufficient to limit the heat loss of the oven to 0.50 K.W. per hour per square foot of oven interior surface when the oven is operating at 260 degrees C. when tested as specified in 4.4.6.

3.6.7 Tolerance and range. The oven temperature shall be automatically maintained within a tolerance of plus or minus 1.5 degrees C. and the range of temperatures throughout the oven shall not exceed 7 degrees C. when set for any temperature between 35 degrees C. and 260 degrees C., when tested as specified in 4.4.7.

3.6.8 Electrical insulation resistance. The electrical insulation resistance of the supports of heating units and other exposed conductor supports shall be at least 50,000 ohms when tested as specified in 4.4.8.

3.6.9 Dielectric strength. The oven shall not break down under the application of the given test potential between line metal parts and dead metal parts, when tested as specified in 4.4.9.

3.6.10 Thermostat operation. The thermostat shall operate when turned on and off and adjusted for various temperature levels without electrical or mechanical failure or undue burning, pitting or welding of the contacts, when tested as specified in 4.4.10.

3.7 Identification marking. Each oven shall be permanently and legibly marked with the name or registered trademark of the manufacturer, serial number, temperature range, wattage, voltage and current type.

3.7.1 Department of Defense marking. In addition to the marking requirements of 3.7, each oven for the military services shall be marked in accordance with MIL-STD-130.



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3.8 Repair parts. The following repair parts shall be furnished, suitably mounted in the electrical wiring compartment.

One complete set of heating elements
One indicator light lamp.
One thermometer.

3.9 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information shall specifically include instructions, drawings, power wiring, structural modifications, and special support devices necessary for roughing-in preparations prior to installation of the unit. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by Military Standard MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located. Service data shall list the repair parts specified in 3.8 and shall indicate where they are stored. When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor



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shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.10 Workmanship. Workmanship shall be first class throughout. Ovens shall be free from defects which detract from their appearance or may impair their serviceability. Welds shall be free of porosity and open seams and shall be ground smooth and polished. Door seal shall be free of frays and bumps and other defects which make the seal discontinuous. Metal edges and corners shall not be rough or sharp.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.



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| | Inspection level | AQL (percent defective) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 2.5 |
| Minor defects | III | 4.0 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

| Component (Unit of product) | Characteristic | Test Procedure | Requirement | Inspection level | AQL |
|---|---|---|---|---|---|
| Shelf | deflection | 4.4.1 | 3.6.1 | S-3 | 1.5 |
| Shelf Supports | rigidity | 4.4.2 | 3.6.2 | S-3 | 1.5 |
| Thermometer | calibration and durability of coloring material | 4.4.3 | 3.6.3 | S-2 | * |
| Oven | 24 hour operation | 4.4.4 Step 4 only | 3.6.4 | S-3 | 1.5 |
| Oven | heating time | 4.4.5 | 3.6.5 | S-4 | * |
| Oven | Heat loss | 4.4.6 | 3.6.6 | S-4 | * |
| Oven | tolerance and range | 4.4.7 | 3.6.7 | S-4 | * |
| Heater supports | insulation resistance | 4.4.8 | 3.6.8 | S-4 | 1.5 |
| Oven | dielectric strength | 4.4.9 | 3.6.9 | S-4 | 1.5 |
| Thermostat | functionability | 4.4.10 | 3.6.10 | S-4 | 2.5 |

* Acceptance number shall be zero.



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4.3 Examination. The ovens shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordanced with the following classification of defects:

Major

101 Component part missing.
102 Oven not free of sharp edges.
103 Welds not free of cracks and porosity.
104 Screws and bolts not secured.
105 Door cannot be closed securely.
106 Thermometer graduations not present, complete, correct, legible, or permanent.
107 Indicator light does not show red when energized
108 Door seal not continuous, and not free of frays and bumps.
109 Seams open and insulation exposed.

Minor

201 Metal surfaces of oven not free of pits and burs.
202 Metal surfaces not free of cracks and scratches.
203 Welds not smoothly polished.
204 Oven not free of dents.
205 Plating does not adhere.
206 Identification marking not present, complete, correct, legible, or permanent.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The ovens shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests.

4.4.1 Shelf deflection test. A shelf shall be supported in the same manner as it is supported in the oven and shall have a load of 50 pounds placed at its center for a period of 5 minutes and then removed to determine conformance to 3.6.1. The base area of the load shall not exceed 30 square inches.



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4.4.2 Shelf support rigidity test. Two shelves, each loaded as described in 4.4.1, shall be placed on the shelf supports in a horizontal position fully inserted in the oven. After 2-1/2 minutes, the shelves loaded as before shall be withdrawn at least one-half the depth of the oven and allowed to remain for an additional 2-1/2 minutes. The shelf supports shall then be examined to determine conformance to 3.6.2.

4.4.3 Thermometer tests. Each thermometer shall be subjected to the calibration and durability of coloring material tests of GG-T-336 for conformance to 3.4.4.

4.4.4 Oven operating test conditions. The following test conditions shall be met before and during conduct of the tests specified in 4.4.5 and 4.4.6.

Step 1. The oven with doors open shall be maintained in an ambient temperature of 25 degrees C. plus or minus 3 degrees C. for 12 hours with oven switch turned "OFF".

Step 2. Two shelves shall be placed in operating position and 7 thermocouples of No. 30 gage (B&S) wire placed at the geometric center of the oven and at points one-fifth of the length of the lines from each end of the three lines connecting the center of the three pairs of opposite sides of the interior of the oven. A wattmeter and voltmeter shall be placed in the electric circuit, the thermometer placed in position, and the doors closed.

Step 3. The ambient temperature throughout the tests shall be maintained at the temperature of step 1 above.

Step 4. With the thermostat set at 260 degrees C., and with exhaust ventilators set at the position specified by the manufacturer for optimum thermal control, the voltage shall be applied by turning the oven switch to the "ON" position.

4.4.5 Heating time test. With the oven meeting the conditions of 4.4.4, compliance with the heating time given in 3.6.5 shall be determined. The arithmetical mean of the seven thermocouples shall be termed the oven temperature.

4.4.6 Heat loss test. Following the test described in 4.4.5 the oven shall continue to be heated for 4 hours, after which the heat loss test shall be made to determine conformance to 3.6.6. The total heat loss shall be determined as being equal to the electrical energy consumed by the oven during the 5th hour of operation. At the end of the 5th hour of operation, thermocouples shall be placed in several positions on all sides of the exterior of the oven to determine surface temperature. A small piece of putty or similar substance shall be used to hold the thermocouples in position against the surface. Power shall continue to be supplied to the oven for 24 hours to fulfill the requirements of 3.6.4.



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4.4.7 Tolerance and range test. The thermostat shall be set successively at even 50 numbered degrees C. At 15-minute intervals, temperature readings shall be taken of all seven thermocouples until equilibrium is reached for the thermostat setting. The range of temperatures throughout the oven and the variation from the equilibrium temperature at each thermocouple shall not exceed the tolerance limits given in 3.6.7. Temperature readings shall be taken five times at five minute intervals following the time of reaching the point of equilibrium at each thermostat setting.

4.4.8 Insulation test. After exposure to air having a relative humidity of 85, plus or minus 5 percent, at a temperature of 32 degrees, plus or minus 2 degrees C., for a period of 24 hours, the insulation resistance of the heating units supports and other exposed conductor supports shall be measured to determine conformance to 3.6.8. A high-resistance voltmeter using a 250-volt direct current circuit shall be used to make the measurement.

4.4.9 Dielectric strength test. The oven shall be tested for dielectric strength to determine conformance to 3.6.9. A 60 cycle alternating potential of 900 volts shall be applied for a period of one minute between live metal parts and dead metal parts with the oven at the operating temperature reached in normal use. The test shall be conducted with a suitable transformer the output voltage of which can be regulated. Starting at zero, the applied potential is to be increased gradually until the required test value is reached or breakdown occurs.

4.4.10 Thermostat test. The thermostat shall be tested to determine conformance to 3.6.10. The thermostat shall successfully complete 50 cycles of operation when the oven is connected to a supply circuit of current type specified in 3.2.2, having a voltage which is 120 percent of the rated voltage of the oven. The thermostat shall be connected so that there will be a difference of potential equal to the test voltage between live metal parts of the control mechanism and the dead metal parts of the oven.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level A.

5.1.1 Unit of issue. One oven, of the type, class, style, and size specified, constitutes one unit of issue.



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5.1.2 Unit preparation. Each unit shall be disassembled as far as is practical. Doors shall be secured. All exterior finished surfaces shall be protected by wrapping with kraft paper or by a kraft paper shroud, and by cushioning with flexible corrugated fiberboard, cellulosic cushioning material, or other suitable material. Wrapping and cushioning material shall be secured in position. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit. Disassembled parts and repair parts shall be packaged in one or more fiberboard cartons or wood boxes of suitable size and design. Containers shall be adequately secured to prevent accidental opening and shall contain sufficient cushioning material to prevent shifting and breakage. Cartons shall be wrapped, and wood boxes shall be lined with waterproofed kraft paper conforming to the requirements of UU-P-271. Carton wraps shall conform to class C-1, and wood box liners shall conform to class H-4, L-2(a), L-2(b), or M-3.

5.2 Packing.

5.2.1 Level A. Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table IV. Oven shall not be packed in an inverted position. The unit shall be secured to the base of the exterior container in such a manner that the weight of the unit shall not be carried directly by its legs. In addition, the unit shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings, to insure against shifting, breakage, or marring or denting or surfaces. All box panels, except the base, shall be lined with waterproof-barrier material conforming to UU-P-271, class M-1, M-2, or M-3, in such a manner as to prevent direct entry of water. As an alternate, each unit shall be protected with a shroud fabricated of material conforming to UU-P-271, class M-1, M-2, or M-3. The exterior container design shall include skids of not less than 1-1/2 inches in thickness, applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. Closure shall comply with appendix of applicable specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.3 Marking.

5.3.1 Interior packages. Each container shall be marked to identify contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Directional arrows in accordance with 5.5.16.3 of MIL-STD-129 shall be applied to indicate correct position for stowing. The legend 'OPEN THIS SIDE' shall be shown, where applicable, to facilitate locating service data, and inspecting and removing the unit.



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6. NOTES

6.1 Intended use. The ovens covered by this specification are intended for use in various heating, drying, baking and dry sterilization procedures in medical laboratories.

6.2 Ordering data. Procurement document should specify the following:

(a) Number and date of this specification.
(b) Style oven required (1.2.1.1).
(c) Size oven required (1.2.1.2).
(d) Level of preservation and packaging required (5.1).
(e) Level of packing required (5.2).
(f) Special marking when required (5.3).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 Supersession data. This specification supersedes in part Federal Specification W-O-830b, in particular the requirements applicable to type I and the following sizes:

| | W-O-830b |
|---|---|
| Size 1 replaces | Size 2 |
| Size 2 replaces | Size 4 |
| Size 3 replaces | Size 7 |

6.5 Laboratory report. The laboratory report is independent proof furnished by a nationally recognized testing agency, other than the contractors own, or affiliated laboratories.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

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